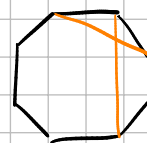
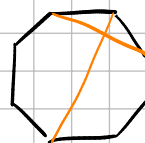
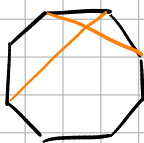
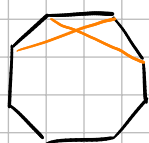
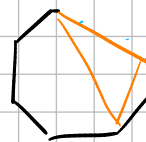
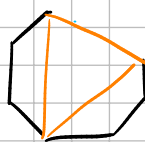
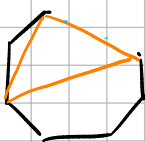
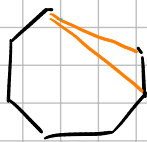
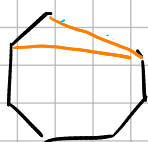
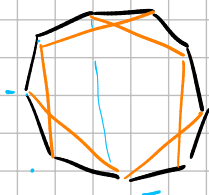
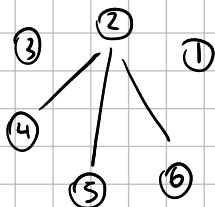
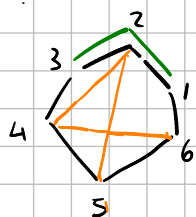


If we select $n-3$ chords (repetitions allowed), for each $k=1, \dots, n-2$ and $i \in \{1, \dots, n\}$ if the vertices $i+1, i+2, \dots, i+k-1$ are only connected within the $(k+1)$ -gon $(i, i+1, \dots, i+k-1)$ then the edges $(i, i+1), (i+1, i+2), \dots, (i+k-1, i+k)$ must be part of a $(k+2)$ -gon (otherwise the term vanishes).

Conjecture: only triangulations survive.

the following happens:

- i connects to $i+k+1$
OR
- $i+k$ connects to $i-1$
OR
- $\begin{cases} i \text{ connects to } i+k+j \\ i+k \text{ connects to } i+k+j \end{cases} \quad j \in \{1, \dots, i-2\}$



$n=4$

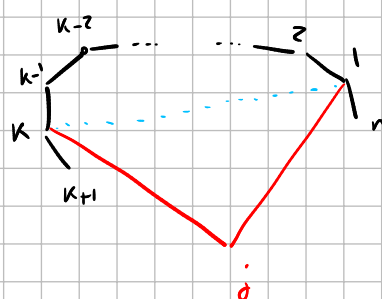
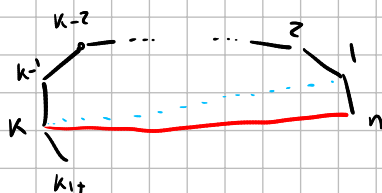
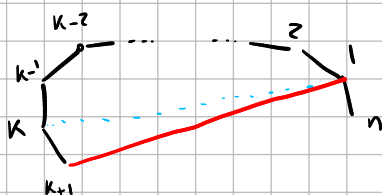


$n=5$

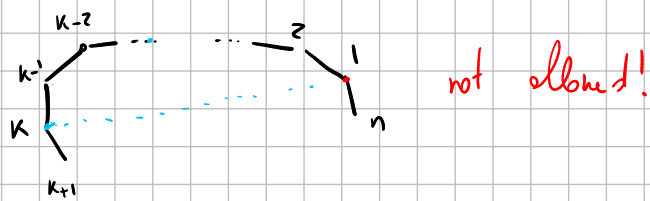
$k \geq 1$: every edge is part of a triangle



assume triangulation and
nothing crosses blue dotted line (not necessarily a chord)



• if K does not connect to anything, must have



not allowed!

→ K must connect

similar idea: z must connect

suppose that we have a triangulation of an n -gon. (with $n-3$ chords)
if we add any chord, it must intersect something.

if nothing crosses X_{ik} , then X_{ik} must be a chord of the triangulation. otherwise, adding X_{ik} would not cross anything.

→ triangulation satisfies our property!

pick $n-3$ chords and draw

1: 3 --- $n-1$

2: 4 --- n

3: 5 --- n

⋮

$n-2$: n

